experiment family	key result (as measured)	sample	test / correction	evidence	verdict
Beacon A10: persistent reuse under information asymmetry	`fas 20/20; direct 0/20; D-noact 0/20; D-flat 20/20 (first action gather\_oak\_log 20/20)`	n=20 x 4 conds	McNemar exact: p=2e-6 (vs direct, noact); p=1.0 (vs D-flat)	L3 behavior	✓ **relational reuse supported; ablation=full**
Write-back isolation (A1_B1 vs A1_B2)	`operation:/change: nodes: 0 vs 13 (identical experience, single switch)`	n=1 lineage (seed 1, bitwise)	deterministic isolation (both completions 5/5 @11)	L1 repro.	✓ **experience -> new structure, not injected**
Cross-context re-activation (G0/G1/G2)	`marked re-lit 4/12 vs 0/8 vs 0/0 (a>=0.05)`	n=5 seeds	deterministic; vs flat-memory control	L1/L2	✓ **episodic structure carries re-activation: access layer**
Functional consumption: ledger + event framework (C, D/C20e/C20g)	`completion 13 vs 16 ticks; first action flip 5/5; real-failure receipts @15 vs @17`	n=5 determinations	deterministic channel isolation	L3 (assembly)	✓ **stored experience changes how (not whether) it acts**
JCG: joint convergence gain (core incremental value)	`JCG>0 in 200/200 paired seeds; >=2-input MRR@1 = 1.0 (prior replication 0.970)`	n=50 x 4 conds	deterministic + prior 0.970 replication	L2 mechanism	✓ **multi-input convergence; mechanism runs**
dJCG vs flat retrieval (preregistered, C-1..C-4)	`median dJCG -0.40/-0.40/-0.78/-0.78; reversals FAS>B2: 0/200`	n=50 paired x 4	1-sided Wilcoxon + Holm: p=1.5e-9; r=0.615; CI excl. 0	L2 mechanism	✗ **H3 rejected: no value over flat**
Shared-hub topology + 84-cell parameter sweep	`FAS 30/30 vs B2 0/30 (w=0.5); sweep: FAS better 11/84 (w<=0.5, d<=1); flat better 9/84 (w>=0.7, d>=1); tie 64`	n=30 seeds x 85 cells	deterministic; design frozen before execution	L2 mechanism	△ **advantage region: weak edges x shallow paths**
Relation directionality (diagnosis, F vs B projection)	`rho(score,act) 0.325 -> 0.645 (I3); reach2 0 -> 9-12; dilution 0.833->0.475`	n=10 matched x 60 runs	Wilcoxon 21 tests: p=0.001953; r=0.886; 10/10 seeds	L2 mechanism	✓* **direction semantics=limit; production untouched**
Explicit resource routing (LLM decision head)	`T4 info-competition 19/20 vs 15/20; raw p=0.046 -> Bonferroni p~1; T3 slower; tokens 2.1-3.7x`	n=20 matched x 520 runs	Wilcoxon; 280 tests Bonferroni: T4 all p~1	L3 exploratory	△ **directional only; no robust value**
Zero-prior starvation (P0 vs P1/P2)	`P0 0/25 with +29 nodes; P1=P2 25/25 @median 4 ticks`	n=25 episodes	deterministic; direct replay diagnosis	L3 boundary	△ **scaffolding is precondition; locks**

Verdicts: ✓ supported as stated; ✗ hypothesis tested and rejected; △ boundary / directional only (✓* = mechanism diagnosis, production graph unmodified).
Effects are per-row incommensurable metrics; only the direction verdict is comparable across rows. Every number is read from the archived run outputs (beacon_v1/analysis_final.json, core_incremental_value/statistics.csv, core_routing_v2 campaign statistics, relation-direction campaign; the prior JCG replication is research_audit/mechanism_falsification).